

Justify all answers. Submit as a single PDF.

1. (10 points) Suppose that Frida selects a ball by first picking one of two boxes at random and then selecting a ball from this box at random. The first box contains two white balls and three blue balls, and the second box contains four white balls and one blue ball. What is the probability that Frida picked a ball from the first box if she has selected a blue ball?
2. Suppose that 4% of the patients tested in a clinic are infected with avian influenza. Furthermore, suppose that when a blood test for avian influenza is given, 97% of the patients infected with avian influenza test positive and that 2% of the patients not infected with avian influenza test positive. What is the probability that
 - (a) (5 points) a patient testing positive for avian influenza with this test is infected with it?
 - (b) (5 points) a patient testing positive for avian influenza with this test is not infected with it?
 - (c) (5 points) a patient testing negative for avian influenza with this test is infected with it?
 - (d) (5 points) a patient testing negative for avian influenza with this test is not infected with it?
3. A space probe near Neptune communicates with Earth using bit strings. Suppose that in its transmissions it sends a 1 one-third of the time and a 0 two-thirds of the time. When a 0 is sent, the probability that it is received correctly is 0.9, and the probability that it is received incorrectly (as a 1) is 0.1. When a 1 is sent, the probability that it is received correctly is 0.8, and the probability that it is received incorrectly (as a 0) is 0.2.
 - (a) (10 points) Find the probability that a 0 is received.
 - (b) (10 points) Use Bayes' theorem to find the probability that a 0 was transmitted, given that a 0 was received.
4. There are three cards in a box. Both sides of one card are black, both sides of one card are red, and the third card has one black side and one red side. We pick a card at random and observe only one side.
 - (a) (5 points) If the side is black, what is the probability that the other side is also black?
 - (b) (5 points) What is the probability that the opposite side is the same color as the one we observed?
5. (10 points) What is the probability that when a fair coin is flipped n times an equal number of heads and tails appear?
6. (10 points) Suppose that A and B are events from a sample space S such that $\Pr(A) \neq 0$ and $\Pr(B) \neq 0$. Show that if $\Pr(B | A) < \Pr(B)$, then $\Pr(A | B) < \Pr(A)$.

7. Suppose that A and B are events with probabilities $\Pr(A) = \frac{3}{4}$ and $\Pr(B) = \frac{1}{3}$.
- (a) (10 points) What is the largest $\Pr(A \cap B)$ can be? What is the smallest it can be? Give examples to show that both extremes for $\Pr(A \cap B)$ are possible.
 - (b) (10 points) What is the largest $\Pr(A \cup B)$ can be? What is the smallest it can be? Give examples to show that both extremes for $\Pr(A \cup B)$ are possible.
8. (10 points) A gambler mixes one cheat die with all sixes into a bag with 8 normal six-sided dice. She randomly draws one from the bag, rolls it twice, and gets six both times. What is the probability she drew the cheat die?
9. (10 points) In a certain country, every family prefers having a boy. Each family repeatedly has children until a boy is born, at which point they stop having more children. Assume the probability of having a boy or a girl is equal, and the gender of each child is independent of previous births. Find the expected ratio of boys to girls in the entire country.